

Cloud Computing

Unit II — Data Storage & Cloud Computing

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Syllabus Coverage

Data Storage: Enterprise Data Storage, DAS, SAN, NAS, Data Storage Management, File System, Cloud Data Stores, Grid Computing. Cloud Storage: Data Management, Cloud Provisioning, Data Intensive Technologies. Cloud File Systems (GFS and HDFS). Cloud Storage from LANs to WANS: Cloud Characteristics, Distributed Data Storage (Amazon Dynamo, CouchDB, ThruDB).

1. Data Storage in Cloud Computing

1.1 Cloud Storage — Introduction

Cloud Storage

Cloud Storage is a service that allows users to save data on an offsite storage system managed by a third-party and made accessible by a web services API.

Cloud storage is also defined as a cloud computing model that stores data on the Internet through a cloud computing provider who manages and operates data storage as a service. Cloud storage is also known as utility storage.

Providers include: Google Drive, Dropbox, OneDrive, iCloud. These provide free storage up to a limit, beyond which users must pay.

1.2 Storage Devices — Classification

Storage devices can be broadly classified into two categories:

Block Storage Devices

Block storage devices offer raw storage to clients. This raw storage is partitioned to create volumes. Used in block-level access protocols.

File Storage Devices

File storage devices offer storage to clients in the form of files, maintaining their own file system. This storage is in the form of Network Attached Storage (NAS).

1.3 Cloud Storage Classes

Cloud storage can be broadly classified into two categories:

Unmanaged Cloud Storage

Unmanaged cloud storage means the storage is preconfigured for the customer. The customer can neither format, nor install his own file system, nor change drive properties.

Managed Cloud Storage

Managed cloud storage offers online storage space on-demand. The managed cloud storage system appears to the user to be a raw disk that the user can partition and format.

1.4 How Cloud Storage Works

Cloud storage works as a virtual data center. It offers end users and applications virtual storage infrastructure that can be scaled to the application's requirements. It generally operates via a web-based API implemented remotely through its interaction with in-house cloud storage infrastructure.

- Cloud storage includes at least one data server to which a user connects via the Internet.
- The user sends files to the data server, which forwards the message to multiple servers, manually or in an automated manner, over the Internet.
- The stored data can then be accessed via a web-based interface.

- Cloud storage is purchased from a third-party vendor who owns and operates data storage capacity and delivers it in a pay-as-you-go model.
- Applications access cloud storage through traditional storage protocols or directly via an API.

1.5 Creating a Cloud Storage System

Cloud providers use the following process to build a cloud storage system:

1. The cloud storage system stores many copies of data on many servers at various locations.
2. The data is stored at various locations so that if one system fails, it can change the pointer location to where the object is stored.
3. The cloud provider uses virtualization software to aggregate storage assets into a cloud storage system. This system is called StorageGRID.
4. StorageGRID creates a virtualization layer which fetches storage from various storage devices into a single management system.
5. It manages the data from CIFS and NFS file systems over the Internet.

1.6 Virtual Storage Container

Virtual Storage Container

Virtual storage containers offer high-performance cloud storage systems. Logical Unit Number (LUN) of device, files, and other objects are created in virtual storage containers. A virtual storage container defines a cloud storage domain.

Challenges of Cloud Storage

Apart from its flexibility and convenience, storing data in the cloud has several challenges. Customers must be able to:

- Get provision for additional storage on-demand.
- Know and restrict the physical location of the stored data.
- Verify how data was erased.
- Have access to a documented process for disposing of data storage hardware.
- Have administrator access control over data.

1.7 Benefits of Cloud Storage

- 1. Cost Saving:** Pay only for storage actually consumed; no upfront hardware investment.
- 2. Data Redundancy and Replication:** Data is replicated across multiple servers and geographic locations, ensuring availability even if one system fails.
- 3. Scalability:** Storage can be scaled up or down on demand without purchasing new hardware.
- 4. Speed:** Optimized delivery infrastructure ensures fast access to stored data from any location.

2. Enterprise Data Storage

2.1 Introduction

Enterprise Storage

"Enterprise storage is a centralized repository for business information that provides common data management, protection."

An Enterprise Storage System is a centralized repository for business information. It provides a common resource for data sharing, management, and protection via connections to other computer systems. Enterprise storage systems are designed to process heavy workloads of business-critical information.

Types of Enterprise Storage:

- Direct-Attached Storage (DAS)
- Network-Attached Storage (NAS)
- Storage Area Networks (SAN)

2.2 Direct Attached Storage (DAS)

DAS

DAS (Direct Attached Storage) is a digital storage device connected directly to the server, workstation, or personal computer via a cable. Applications use the block-level access protocol for accessing the data. DAS is not a part of the storage network — no network is needed to attach the device to the server.

Examples of DAS devices: Solid-State Drives (SSDs), Hard Drives, Tape Libraries, Optical Disk Drives. The DAS system is attached directly to the computer through the HBA (Host Bus Adapter). Modern DAS systems include integrated disk array controllers with advanced features.

Types of DAS

Internal DAS

The storage device is attached internally to the server or PC by the HBA. HBA is used for high-speed bus connectivity over a short distance.

External DAS

The external storage device is directly connected to the server without any intermediate network device. FCP and SCSI are the protocols acting as an interface between server and storage device.

2.3 Network Attached Storage (NAS)

NAS

Network-Attached Storage (NAS) is a file storage device connected to the network that enables multiple users to access data from the centralized disk capacity. Users on a LAN access the shared storage by the Ethernet connection. This storage is fast, low-cost, and offers all the advantages of a public cloud on-site.

NAS uses file access protocols such as NFS (Unix systems), SMB/Server Message Block (Windows systems), NCP, or AFP (Apple computers). It is designed for network systems processing millions of operations per minute. It is more economical than file servers and more versatile than external disks.

Advantages of NAS

- Easy to install and configure — simple NAS architecture.
- Every user or client in the network can easily access NAS.
- More reliable than simple hard disks.
- Good performance in serving files.
- NAS devices are scalable and can be easily accessed remotely.

Disadvantages of NAS

- Data transfer speed is not as fast as DAS.
- Users require basic knowledge of computer networks to use NAS efficiently.
- Backup cannot be done directly; it requires the installed operating system.

2.4 Storage Area Network (SAN)

SAN	<i>Storage Area Network (SAN) is a dedicated, specialized, and high-speed network which provides block-level data storage. It delivers the shared pool of storage devices to more than one server. The main aim of SAN is to transfer the data between the server and storage device. It is a dedicated network — not accessible through the LAN.</i>
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SAN consists of hosts, switches, and storage devices interconnected using topologies, protocols, and technologies. SAN is mainly used for accessing storage devices such as tape libraries and disk-based devices from servers.

SAN Protocols

FCP (Fibre Channel Protocol)	<i>The most commonly used protocol of SAN. It is a mapping of SCSI commands over the Fibre Channel (FC) network.</i>
iSCSI (Internet SCSI)	<i>Internet Small Computer System Interface. The second-largest block/SAN protocol. It puts SCSI commands inside an Ethernet frame and transports them over an Internet Protocol (IP) Ethernet.</i>
FCoE (Fibre Channel over Ethernet)	<i>Fibre Channel Over Ethernet — similar to iSCSI. It puts the Fibre Channel inside the Ethernet datagram and transports it over an IP Ethernet network.</i>
NVMe (Non-Volatile Memory Express)	<i>A SAN protocol that accesses flash storage via the PCI Express bus.</i>

Advantages of SAN

- Highly scalable.
- Strong security — users can easily implement various security measures.
- Storage devices can be easily added or removed from the network.
- Low cost compared to other storage networks.
- Better disk utilization.

2.5 Comparison Tables

NAS vs. DAS

Parameter	NAS	DAS
Full Form	Network Attached Storage	Direct Attached Storage
Backup/Recovery	Uses files for backup and recovery	Uses sectors for backup and recovery
Complexity	Complex	Simple
Setup	Slightly difficult to set up	Easy to set up and install
Cost	Higher than DAS	Low cost
Capacity	10 ⁹ to 10 ¹² bytes	Only 10 ⁹ bytes
File Sharing (cross-OS)	Allows sharing across different OS	Does not allow sharing across OS
Transmission	Ethernet and TCP/IP	IDE/SCSI

DAS vs. SAN

Parameter	DAS	SAN
Full Form	Direct Attached Storage	Storage Area Network
Backup/Recovery	Uses sectors for backup/recovery	Block by block copying technique
Complexity	Simple	Complex
Setup	Easy to set up and install	Difficult to install and set up
Cost	Low	Higher than DAS
Capacity	Only 10 ⁹ bytes	More than 10 ¹² bytes
File Sharing (cross-OS)	Does not allow sharing across OS	Allows sharing across OS
Transmission	IDE/SCSI	Internet Protocol and Fibre Channel

3. Data Storage Management

3.1 Definition

Cloud Data Management

Cloud data management is a way to manage data across cloud platforms, either with or instead of on-premises storage. The cloud is useful as a data storage tier for disaster recovery, backup, and long-term archiving. With cloud data management, resources can be purchased as needed.

Storage management ensures data is available to users when they need it. Data storage management is typically part of the storage administrator's job. Organizations without a dedicated storage administrator might use an IT generalist for storage management.

Cloud data management is the practice of storing a company's data at an offsite data center typically owned by vendors such as AWS or Microsoft Azure. Managing data in the cloud provides an automated backup strategy, professional support, and ease of access from any location.

3.2 Benefits of Cloud Data Management

- 1. Security:** Modern cloud data management delivers better data protection than on-premises solutions. 94% of cloud adopters report security improvements. Cloud reduces data loss risk from device damage or hardware failure. Specialized cloud hosting companies employ more advanced security measures than most organizations can achieve on-premises.
- 2. Scalability and Savings:** Cloud data management lets users scale services up or down as needed. More storage or compute power can be added for changing workloads. Companies can scale back after a big project to avoid paying for unused services.
- 3. Governed Access:** Team members can access the data they need from wherever they are. This supports a collaborative work culture — employees can work together on datasets, share insights, and more.
- 4. Automated Backups and Disaster Recovery:** The cloud vendor manages and automates data backups so the company can focus on other activities. An up-to-date backup at all times speeds up disaster recovery and mitigates ransomware attack effects.
- 5. Improved Data Quality:** An integrated, well-governed cloud data management solution tears down data silos and creates a single source of truth for every data point. Data remains clean, consistent, up-to-date, and accessible for real-time analytics, ML applications, and external sharing via APIs.
- 6. Automated Updates:** Cloud providers run application updates automatically. Teams don't need to pause work while waiting for IT to update systems.

4. File System in Cloud

4.1 Definition

Cloud File System

A file system in the cloud is a hierarchical storage system that provides shared access to file data. Users can create, delete, modify, read, and write files and can organize them logically in directory trees for intuitive access.

The vendor creates a file system that offers traditional file protocols like NFS or SMB to cloud-hosted applications. Essentially, the vendor provides an instance of their file system and the organization implements it in the cloud provider of their choice. It then allocates the appropriate storage compute performance and storage I/O.

The goal with cloud file systems is to speed the migration of applications to the cloud. By using a file system in the cloud, the organization does not need to re-write the storage I/O components of the application.

4.2 Google File System (GFS)

Exam Tip

GFS and HDFS are listed in the syllabus as key cloud file systems. Know their key design principles and architecture.

GFS

Google File System (GFS) is a scalable distributed file system developed by Google for large distributed data-intensive applications. It provides fault tolerance while running on inexpensive commodity hardware, and delivers high aggregate performance to a large number of clients.

Key Characteristics of GFS

- Designed for large files — typical files are multi-GB in size.
- Files are divided into fixed-size chunks (64 MB each) stored on chunk servers.
- A single master server manages metadata; multiple chunk servers store actual data.
- Data is replicated across three or more chunk servers for fault tolerance.
- Optimized for sequential read workloads (streaming reads), not random access.
- Supports concurrent appends from multiple clients (record append operation).
- The master maintains: namespace (file names), chunk locations, access control info.

GFS Architecture

- Master Node: Manages file system metadata — namespace, chunk mapping, locations. Communicates with all chunk servers through heartbeat messages.
- Chunk Servers: Store actual data in fixed-size 64 MB chunks on local disks. Each chunk is replicated (default: 3 replicas) across different chunk servers.

- Clients: Access data directly from chunk servers after getting chunk location from the master. Clients do not cache data through the master.

4.3 Hadoop Distributed File System (HDFS)

HDFS

Hadoop Distributed File System (HDFS) is an open-source distributed file system designed to run on commodity hardware. It is highly fault-tolerant and is designed to be deployed on low-cost hardware. HDFS provides high-throughput access to application data and is suitable for applications with large data sets.

Key Characteristics of HDFS

- Handles very large files — designed for files of GB to TB size.
- Write-once, read-many access model — optimized for streaming data access.
- Files are divided into blocks (default: 128 MB) stored across DataNodes.
- A single NameNode manages the file system namespace and metadata.
- Data blocks are replicated across multiple DataNodes (default replication factor: 3).
- Works on commodity (inexpensive) hardware — built for hardware failure tolerance.
- Rack-aware replication: replicas are distributed across different racks for fault tolerance.

HDFS Architecture

- NameNode: The master server that manages the file system namespace and regulates access to files by clients. Stores all metadata in memory.
- DataNodes: Manage storage attached to the nodes they run on. They serve read and write requests from the file system's clients and perform block creation, deletion, and replication on instruction from the NameNode.
- Secondary NameNode: Performs periodic checkpoints of the NameNode — NOT a backup of NameNode.

GFS vs. HDFS Comparison

Parameter	GFS	HDFS
Developed By	Google	Apache (open-source)
Chunk/Block Size	64 MB chunks	128 MB blocks (default)
Master Node	Single Master	NameNode
Storage Nodes	Chunk Servers	DataNodes
Replication	3 replicas (default)	3 replicas (default)
Access Pattern	Sequential (streaming reads)	Write-once, read-many
Availability	Proprietary (Google internal)	Open-source (Apache Hadoop)
Use Case	Google internal apps (web index, etc.)	Big Data processing (MapReduce, Spark)

5. Cloud Data Stores

5.1 Introduction

Cloud Data Stores

Cloud storage is a cloud computing model that stores data on the Internet through a cloud computing provider who manages and operates data storage as a service. The biggest advantage of cloud storage is the ability to store any type of data in digital form and access it from anywhere, anytime, on any device.

Applications access cloud storage through traditional storage protocols or directly via an API. Many vendors offer complementary services designed to help collect, manage, secure, and analyze data at massive scale.

5.2 Cloud vs. Grid Computing

Parameter	Cloud Computing	Grid Computing
Architecture	Client-server computing architecture	Distributed computing architecture
Scalability	High scalability	Normal scalability
Flexibility	More flexible	Less flexible
Management	Centralized management	Decentralized management
Ownership	Cloud servers owned by infrastructure providers	Grids owned and managed by the organization
Services Used	IaaS, PaaS, SaaS	Distributed computing, distributed information, distributed pervasive
Orientation	Service-oriented	Application-oriented
Access	Standard web protocols	Grid middleware

6. Cloud Provisioning

6.1 Definition

Cloud Provisioning

Cloud provisioning means allocating a cloud service provider's resources to a customer. It is a key feature of cloud computing. It refers to how a client gets cloud services and resources from a provider. The cloud services customers can subscribe to include IaaS, SaaS, and PaaS in public or private environments.

6.2 Benefits of Cloud Provisioning

- **Scalability:** In the cloud provisioning model, cloud resources can scale up and scale down entirely dependent on short-term consumption. Traditional IT provisioning requires huge upfront infrastructure investment and capacity forecasting.
- **Speed:** Developers can schedule jobs themselves, removing the need for an administrator who provisions and manages resources. This accelerates delivery.
- **Cost Savings:** Traditional technology incurs huge costs while cloud providers allow customers to pay only for what they consume — a major reason cloud provisioning is preferred.

6.3 Types of Cloud Provisioning

Network Provisioning

In the telecom industry, network provisioning refers to the provisions of telecommunications services to a client.

Server Provisioning

Involves the datacenter's physical infrastructure, installation, configuration of software, and linking it to middleware, networks, and storage.

User Provisioning

A method of identity management that helps keep a check on access and privileges of authorization. Featured by artifacts such as equipment and suppliers.

Service Provisioning

Requires setting up a service and handling its related data.

6.4 Tools Used in Cloud Provisioning

Public cloud providers offer automated tools to provision resources. Major tools include:

- IBM Cloud Orchestrator
- Cloud Bolt
- Morpheus Data
- Flexera
- Cloud Sphere
- Scalr
- Google Cloud Deployment Manager

7. Data Intensive Technologies for Cloud Computing

7.1 Definition

Data Intensive Computing

Data Intensive Computing is a class of parallel computing which uses data parallelism in order to process large volumes of data. The size of this data is typically in terabytes or petabytes. This large amount of data generated each day is referred to as Big Data.

7.2 Characteristics of Data Intensive Computing

- 1. Minimize Data Movement:** To achieve high performance, it is necessary to minimize the movement of data. This reduces system overhead and increases performance by allowing algorithms to execute on the node where the data resides.
- 2. Machine-Independent Approach:** The data intensive computing system utilizes a machine-independent approach where the runtime system controls the scheduling, execution, load balancing, communications, and movement of programs.
- 3. Focus on Reliability and Availability:** Traditional large-scale systems may be susceptible to hardware failures, communication errors, and software bugs. Data intensive computing is designed to overcome these challenges with high reliability and availability.
- 4. Designed for Scalability:** Data intensive computing can accommodate any amount of data and meet time-critical requirements. Scalability of both hardware and software architecture is one of its biggest advantages.

8. Cloud Storage from LANs to WANs

8.1 Cloud Characteristics (LAN to WAN Context)

Characteristic 1: Elastic Computing Power

Computer power is elastic when it can perform parallel operations. Applications conceived to run on a shared-nothing architecture are well matched for such an environment. For example, Google's App Engine supplies not only a cloud computing infrastructure, but also an entire program stack with a constrained API so that software developers are compelled to compose programs that can run in a shared-nothing environment — thereby enabling elastic scaling.

Characteristic 2: Data Stored at Unknown Host

In general, letting go of data is a threat to many security issues and suitable precautions should be taken. Cloud computing implies that computing and storage resources are being operated from an unknown position. The data is physically stored in a specific host country and is subject to localized laws and regulations. Since most cloud computing vendors give their clients little control over where data is stored, the client has no alternative but to expect the data is encrypted using a key unavailable with the owner, and that the data may be accessed by a third party without the customer's knowledge.

Characteristic 3: Data Duplicated Across Distant Locations

Data accessibility and durability are paramount for cloud storage providers, as data tampering can be damaging for both the business and the organization's reputation. Data accessibility and durability are accomplished through hidden replications. Large cloud computing providers with data hubs dispersed worldwide provide high levels of error resistance by duplicating data at distant locations across continents. Amazon S3 replicates data over 'regions' and 'availability zones' so data and applications can survive even when the whole location collapses.

8.2 Distributed Data Storage

Distributed Data Storage

Distributed storage is evolving from existing data storage practices for new-generation WWW applications through organizations like Google, Amazon, and Yahoo. Distributed storage is preferred over traditional relational database systems for reasons of scalability, accessibility, and performance. New-generation applications require processing data to the tune of terabytes and petabytes.

Emerging distributed data storage solutions include: Amazon Dynamo, CouchDB, and ThruDB.

8.3 Amazon Dynamo

Amazon Dynamo

Amazon Dynamo is a widely used key-value store and one of the main components of Amazon.com, the biggest e-commerce store in the world. It has a primary-key only interface — data is retained as key-value pairs, and the only interface to access data is by identifying the key. Values are anticipated to be small (less than 1 MB).

- Highly accessible for writing (composing) as opposed to reading.
- Malfunction of writing inconveniences the end-user more, so it is optimized for writes.
- Data conflicts (confrontations) are resolved at the time of reading rather than writing.

8.4 CouchDB

CouchDB

CouchDB is a document-oriented database server, accessible by REST APIs. 'Couch' is an acronym for 'Cluster Of Unreliable Commodity Hardware', emphasizing the distributed environment. CouchDB is designed for document-oriented applications such as forums, bug tracking, wikis, and Internet notes. It is ad-hoc and schema-free with a flat address space.

CouchDB's Four Pillars of Data Management

- **Save:** ACID compliant — saves data efficiently and reliably.
- **See:** Easy retrieval, straightforward querying procedures, full-text search.
- **Secure:** Strong compartmentalization, ACL (Access Control Lists), connections over SSL.
- **Share:** Distributed means — data can be replicated across servers.

Key Technical Features of CouchDB

- **Storage Model:** Multiversion Concurrency Control (MVCC) scheme with optimistic locking. A client sees a snapshot of the data and works with it even if it is altered simultaneously by a different client.
- **Authentication:** CouchDB has no separate authentication scheme — it is built-in.
- **Replication:** Distributed replication — a server can update others once it comes back online after being offline. If there are conflicts, CouchDB chooses a survivor and holds it as the latest revision.
- **Conflict Resolution:** Yields identical results regardless of the order of offline revisions — ensures data consistency.

8.5 ThruDB

ThruDB

ThruDB aspires to be universal in simplifying the administration of the up-to-date WWW data layer (indexing, caching, replication, backup) by supplying a reliable set of services.

ThruDB Services

- **Thrucene:** For indexing.
- **Throxy:** For partitioning and load balancing.
- **Thrudoc:** For article/document storage.

ThruDB Foundation — Open Source Projects

- Thrift — cross-language data serialization, RPC, and server programming.
- Lucene — indexing.
- Spread — message bus.
- Memcached — caching.
- Backup — backup to disk/S3.
- Amazon S3 — cloud storage backend.

Apache Thrift

Thrift is a framework for effective cross-language data serialization, RPC, and server programming. It is a programs library and set of code-generation devices conceived to expedite development and implementation of effective and scalable backend services. Thrift permits developers to define data types and service interfaces in a single language-neutral document and generate all the necessary code to build RPC clients and servers.

ThruDB uses two data storage engines — Disk and S3. Data is persisted on local computer disk (providing high throughput capability), and a slave process replays all instructions to the S3 backend as well — giving a fault-free persistence and recovery mechanism for virtual environments such as EC2.

9. Quick Revision Glossary

Exam Strategy

All definitions here carry keyword marks. Write them exactly as given during exams, then elaborate in your own words for full marks.

Cloud Storage	<i>A service that allows saving data on an offsite storage system managed by a third-party, accessible via a web services API.</i>
Enterprise Storage	<i>A centralized repository for business information that provides common data management and protection.</i>
DAS	<i>Digital storage device connected directly to the server/workstation/PC via a cable. Uses block-level access protocol. Not part of the storage network.</i>
NAS	<i>File storage device connected to the network, enabling multiple users to access data from centralized disk capacity via Ethernet. Uses NFS, SMB, AFP protocols.</i>
SAN	<i>Dedicated, high-speed network providing block-level data storage. Delivers a shared pool of storage to multiple servers. Uses FCP, iSCSI, FCoE, NVMe protocols.</i>
GFS	<i>Google File System — scalable distributed file system for large data-intensive apps. Files split into 64 MB chunks; managed by a single Master; replicated across Chunk Servers.</i>
HDFS	<i>Hadoop Distributed File System — open-source distributed file system on commodity hardware. Files split into 128 MB blocks; managed by NameNode; stored on DataNodes.</i>
Cloud Data Management	<i>Way to manage data across cloud platforms. Useful for disaster recovery, backup, and long-term archiving. Resources purchased as needed.</i>
Cloud Provisioning	<i>Allocating a cloud service provider's resources to a customer. Includes Network, Server, User, and Service provisioning.</i>
Data Intensive Computing	<i>Class of parallel computing using data parallelism to process large volumes of data (terabytes/petabytes). Related to Big Data.</i>
Unmanaged Cloud Storage	<i>Preconfigured cloud storage where the customer cannot format, install file systems, or change drive properties.</i>
Managed Cloud Storage	<i>On-demand cloud storage that appears to the user as a raw disk — the user can partition and format it.</i>
StorageGRID	<i>A cloud provider's virtualization software that aggregates storage assets from various storage devices into a single management system.</i>
Amazon Dynamo	<i>Widely used key-value store by Amazon. Primary-key only interface; highly optimized for write operations; conflicts resolved at read time.</i>

CouchDB	<i>Document-oriented database server accessible by REST APIs. Cluster Of Unreliable Commodity Hardware. Schema-free; uses MVCC; supports distributed replication.</i>
ThruDB	<i>WWW data layer administration system providing indexing (Thrucene), load balancing (Throxy), and document storage (Thrudoc). Built on Thrift, Lucene, Memcached, S3.</i>
FCP	<i>Fibre Channel Protocol — most common SAN protocol; maps SCSI commands over Fibre Channel network.</i>
iSCSI	<i>Internet SCSI — puts SCSI commands inside Ethernet frames; transports over IP Ethernet. Second-largest SAN protocol.</i>
MVCC	<i>Multiversion Concurrency Control — a storage scheme (used in CouchDB) where each client sees a snapshot of the data even if it is being simultaneously modified by another client.</i>

10. Probable Exam Questions

Note

These are high-probability questions based on syllabus weightage and slide content. All answers are covered in the notes above.

2-Mark / Short Answer Questions

- Define Cloud Storage.
- Define Enterprise Storage.
- What is DAS? Name its two types.
- What is StorageGRID?
- Define Cloud Provisioning.
- What is Data Intensive Computing?
- What does HDFS stand for? Who developed it?
- What is CouchDB? What does COUCH stand for?
- List the four SAN protocols.
- What is MVCC in CouchDB?

5-Mark / Medium Answer Questions

- Explain the two types of cloud storage classes — Managed and Unmanaged.
- Explain NAS with advantages and disadvantages.
- Explain SAN with its four protocols and advantages.
- Compare NAS vs. DAS (with table).
- Compare DAS vs. SAN (with table).
- Explain the process of creating a cloud storage system. What is StorageGRID?
- Explain the benefits of Cloud Data Management (any four).
- Explain the types of Cloud Provisioning.
- Explain the characteristics of Data Intensive Computing.
- Compare Cloud Computing vs. Grid Computing (with table).

10-Mark / Long Answer Questions

- Explain DAS, NAS, and SAN in detail. Compare all three with a table.
- Explain GFS and HDFS. Compare them with a table.
- Explain Amazon Dynamo, CouchDB, and ThruDB as distributed data storage solutions.
- Explain cloud storage from LANs to WANs — characteristics and distributed storage.
- Explain Data Storage Management in Cloud Computing with its six benefits.
- Explain Cloud Provisioning in detail — definition, benefits, types, and tools.